

MISSION 04 · MATTER

How does a 100 000-tonne ship float?

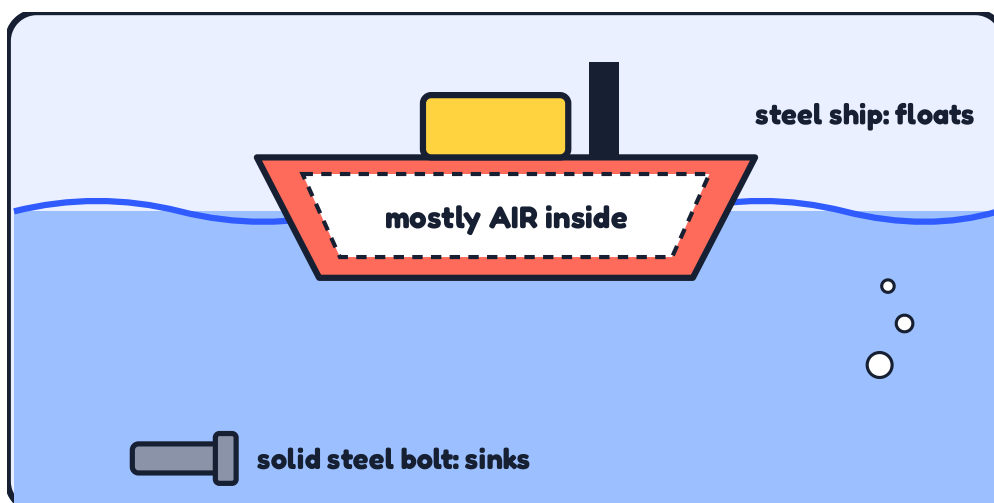
A steel bolt drops straight to the bottom of the sea. A steel ship the size of a shopping mall floats. Same metal, same sea. By the end of this mission you'll know exactly why, and you'll be able to prove it with numbers.

Sec 1-2

5 stages

100 XP to earn

3 badges



MISSION GOALS · BY THE END YOU CAN

1. explain what **mass** and **volume** mean, with their units
2. calculate **density** using $\rho = m / V$
3. measure the volume of odd-shaped objects by **displacement**
4. predict whether something will **float or sink**

Name _____

Class _____

Date _____

tracker

YOUR MISSION ROUTE

- 1 Predict 2 Mass & volume 3 Density 4 Measure it
5 Float or sink?

COLOUR IN YOUR XP

10	20	30	40	50	60	70	80	90	100
----	----	----	----	----	----	----	----	----	-----

☐ Predictor	☐ Card collector	☐ Density detective
---------------------------------	--------------------------------------	---

STAGE

01

+10 XP

Predict 1

Answer in the Annex.
Guess first!

heading

Predict

Make a guess before you learn anything. Being wrong is fine!

predict

PREDICT FIRST · NO CALCULATOR, JUST YOUR GUT

A steel bolt sinks. A steel ship floats. What's different?

- (A) The ship's steel is a lighter kind of steel
(B) The ship is mostly air inside
(C) Salt water holds up big things but not small things

Why do you think so?

STAGE

02

? What is mass?

? What is
volume?

? How many cm^3
is 1 mL?

heading

Mass and volume

Two things you need to know about any object.

concept

Mass is how much matter (stuff) is in an object. We measure it in **grams (g)** or **kilograms (kg)** using an electronic balance.

Volume is how much space an object takes up. We measure it in **cubic centimetres (cm^3)** or **cubic metres (m^3)**. For liquids we often use millilitres, and **1 mL = 1 cm^3** .

STAGE

03

? Define density in your own words.

Same volume, different mass

- A 1 cm³ cube of cork: 0.24 g
- A 1 cm³ cube of water: 1 g
- A 1 cm³ cube of steel: 7.8 g

Same mass, different volume

- 1 kg of feathers fills a pillowcase
- 1 kg of water fills a 1 L bottle
- 1 kg of gold is smaller than a phone

compare

Density

How tightly the stuff is packed.

heading

Density tells you how much mass is packed into each cm³ of a material.

concept

Two things made of the same material have the same density, however big or small they are. A steel paper clip and a steel ship's plate have the same density.

$$\rho = m / V$$

 ρ density (say "rho") g/cm³ m mass g V volume cm³

formula

Memory trick: density = mass ÷ volume. Mass goes on top because it is the "stuff".

+10 XP

COLLECT ALL FOUR · READ THEM, THEN TEST A FRIEND

cards

CORK

0.24g/cm³

So light it floats on almost any liquid. Used for fishing floats.

ICE

0.92g/cm³

Less dense than water, so icebergs float, with most of their bulk underwater.

WATER

1.00g/cm³

The benchmark. Denser than 1 sinks in water; less than 1 floats.

GOLD

19.3g/cm³

A gold bar the size of a phone weighs more than a big bag of rice.

+10 XP

Worked example 1

[2]

Try it in the box. Full solution in the Annex.

worked-example

A block measures 2 cm × 3 cm × 5 cm. Its mass is 81 g. Find its density.

+10 XP

Error hunt 1

Someone else's working. Find the slip.

Jia Hui worked out the density of a 60 g rubber ball with a volume of 50 cm³.

error-hunt

L1 $\rho = V / m$

L2 $\rho = 50 / 60$

L3 $\rho = 0.83 \text{ g/cm}^3$, so it floats

Circle the line with the mistake. Then write the correct working below.

STAGE

04

Measure it

heading

What if the object is a weird shape?

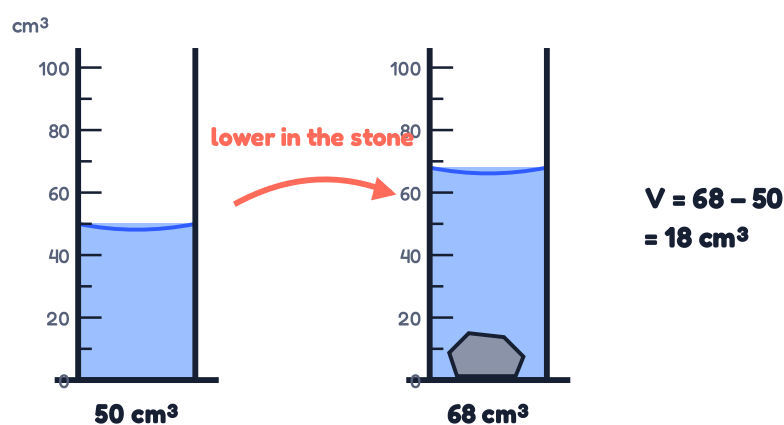
? How do you find the volume of a stone?

For a box shape, multiply **length × breadth × height**. A stone has no neat sides, so we use the **displacement method**:

concept

1. Pour water into a measuring cylinder. Read the volume at the bottom of the curved surface (the meniscus), with your eye level with it.
2. Tie the stone to a thread and lower it in gently, so no water splashes out.
3. Read the new volume.
4. Volume of the stone = new reading – first reading.

figure



The water rises by exactly the stone's volume, because the stone pushes that much water out of the way.

+10 XP

Q1

● ○ ○

Warm-up

[3]

question

A stone has a mass of 54 g. It is lowered into a measuring cylinder, and the water level rises from 50 cm³ to 70 cm³.

(a) Find the volume of the stone.

[1]

(b) Find the density of the stone.

[2]

STAGE

05

? When does an object float?

Float or sink?

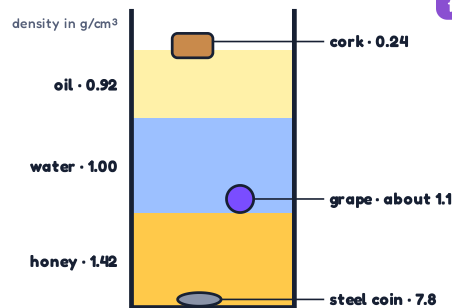
heading

The rule that explains the ship.

concept

Compare the density of the object with the density of the liquid.

- Object **less dense** than the liquid → it **floats**.
- Object **more dense** than the liquid → it **sinks**.
- Liquids stack up the same way: the densest liquid sinks to the bottom.



figure

DID YOU KNOW?

Dead Sea

The Dead Sea is so salty (density about 1.24 g/cm³) that people float on it easily while reading a newspaper. Your body is about 1.0 g/cm³.

row fact

Each object sinks until it reaches a layer denser than itself.

An object has a mass of 30 g and a volume of 40 cm³. What happens when you drop it in water?

mcq

- (A) It sinks: density 1.33 g/cm³ (B) It floats: density 0.75 g/cm³
 (C) It sinks: density 0.75 g/cm³ (D) It floats: density 1.33 g/cm³

+10 XP

Q2

Multiple choice. Tick one box.

[1]

+10 XP

Activity 1

activity

TRY IT AT HOME · WITH AN ADULT

Rainbow in a glass

You need: a tall clear glass, honey or syrup, water with food colouring, cooking oil, a grape, a small cork

1. Pour in a layer of honey. Then slowly pour coloured water down the side of the glass.
2. Slowly pour in the oil last.
3. Drop in the grape and the cork. Watch where each one stops.

Explain where the grape stopped, using the word density.

+10 XP

STICKY-NOTE CHALLENGE · DO ALL THREE IN YOUR BOOK

stickies

Draw

Draw the ship from the cover. Label the steel and the air, and write "average density < 1.03 g/cm³".

Explain

In one sentence, tell a Primary 5 kid why a life jacket helps you float.

Challenge

Ice is 0.92 g/cm³. Sea water is 1.03 g/cm³. Will an ice cube float higher in sea water or in tap water?

summary

MISSION DEBRIEF · EXPLAIN HOW A STEEL SHIP FLOATS, USING THE WORD DENSITY

HOW DO YOU FEEL ABOUT DENSITY? TICK ONE

confidence

☐ **Still wobbly**

Re-read stage 3 and redo the worked example.

Review on ____ / ____

☐ **Getting it**

Redo the error hunt and the stone question.

Review on ____ / ____

☐ **Nailed it**

Try the sticky-note challenge on a friend.

Review on ____ / ____

STAGE

A

Answers · check your mission

Only look after you have tried. Colour in the XP you earned on the tracker at the start.

Predict 1

1 · Predict

The ship is mostly air inside. Both are made of the same steel, which is about 8 times denser than water.

But the hull is hollow. Steel plus all that air has an average density lower than sea water, so the ship floats. Fill it with water (a hole in the hull) and it sinks.

Worked example 1

3 · Density

1 Work out the volume first.

$$V = 2 \times 3 \times 5 = 30 \text{ cm}^3$$

2 Divide mass by volume.

$$\rho = 81 \div 30 = 2.7 \text{ g/cm}^3$$

2.7 g/cm³ is the density of aluminium, so the block is probably aluminium.

Answer: **2.7 g/cm³**

Error hunt 1

L1 is wrong. It should be mass ÷ volume, not volume ÷ mass.

$\rho = 60 \div 50 = 1.2 \text{ g/cm}^3$. That is more than 1, so the ball **sinks**.

Tip: say it out loud, “density equals mass over volume”, before you write anything.

4 · Measure it

Q1

(a)

$$V = 70 - 50 = 20 \text{ cm}^3$$

(b)

$$\rho = 54 \div 20 = 2.7 \text{ g/cm}^3$$

5 · Float or sink?

B

$\rho = 30 \div 40 = 0.75 \text{ g/cm}^3$. That is less than water (1.00), so it floats.

Q2 · MCQ

Activity 1

The grape (about 1.1 g/cm³) is **denser than oil and water**, so it sinks through them.

It is **less dense than honey** (1.42 g/cm³), so it stops on top of the honey layer.

**Sticky note ·
Explain**

A life jacket is full of foam or air, which has a very low density, so it lowers your **average density** below that of water and you float.

**Sticky note ·
Challenge**

Higher in sea water. Sea water is denser, so less of the ice needs to be under the surface to hold it up.